

In *Who's who* his recreations were given as "travelling and the usual school games" until he was sixty. On his 60th birthday he competed in a seven-mile cross country run of the Wirral Athletic Club; thereafter his sole recreation was "travelling". He had also a remarkable private library, and was an authority on local history in the Wirral. He was a schoolmaster all his working life—at Gresham's School, Holt and at Rossall and Solihull and (1927–30) in Natal; from 1930 to 1946 he was headmaster of Birkenhead School. He was for many years a staunch supporter of the Liverpool Mathematical Society (later a branch of the M.A.). To be driven back to Birkenhead by him through the Mersey tunnel after attending a meeting of that society was a memorable experience—not least for the running commentary maintained in his great voice above the roar of the traffic.

The photograph of W. F. Bushell in *Gazette* 31 suggests a bulldog personality but less clearly his large-heartedness, loyalty and uninhibited friendliness. His last days were much marred by rheumatism and arthritis which rendered him almost immobile; they were relieved by the care and attention of his devoted housekeeper, Mary, until about two years ago when her own health compelled her to make way for another no less careful of his comfort.

I am indebted to Professor A. G. Walker, F.R.S. and to two other members of our Liverpool branch for help in preparing this obituary notice.

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Notes

59.3 The irrationality of $\sqrt{2}$

The following proof of the irrationality of $\sqrt{2}$ seems to me simpler than that which is usually attributed to Pythagoras.

Let S be the set of those natural numbers n for which $n\sqrt{2}$ is an integer. If S were not empty, it would have a least element k , say. Consider the number $(\sqrt{2} - 1)k$. Then

$$(\sqrt{2} - 1)k\sqrt{2} = 2k - k\sqrt{2},$$

and, since $k \in S$, both $(\sqrt{2} - 1)k$ and $2k - k\sqrt{2}$ are natural numbers. So, by definition $(\sqrt{2} - 1)k \in S$. But $(\sqrt{2} - 1)k < k$, contradicting the assumption that k is the least element of S . Hence S is empty, which means that $\sqrt{2}$ is irrational.

If in this proof we replace 2 by any natural number h whose square root is not an integer, and replace 1 by the greatest integer less than $\sqrt{h}$, we obtain a proof that the square root of any natural number is either an integer or irrational.

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